

PhumPhae KrungThep

WEGO

– Weather Effect Going Outside –

Air Pollution - PM2.5



Hot Weather



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Presentation Outline



- Overview and Objectives
- Overall Architecture
- Data Collection
- Database schema and Integration
- Data Sharing API
- Data Visualization
- Key Feature

Overview and Objectives



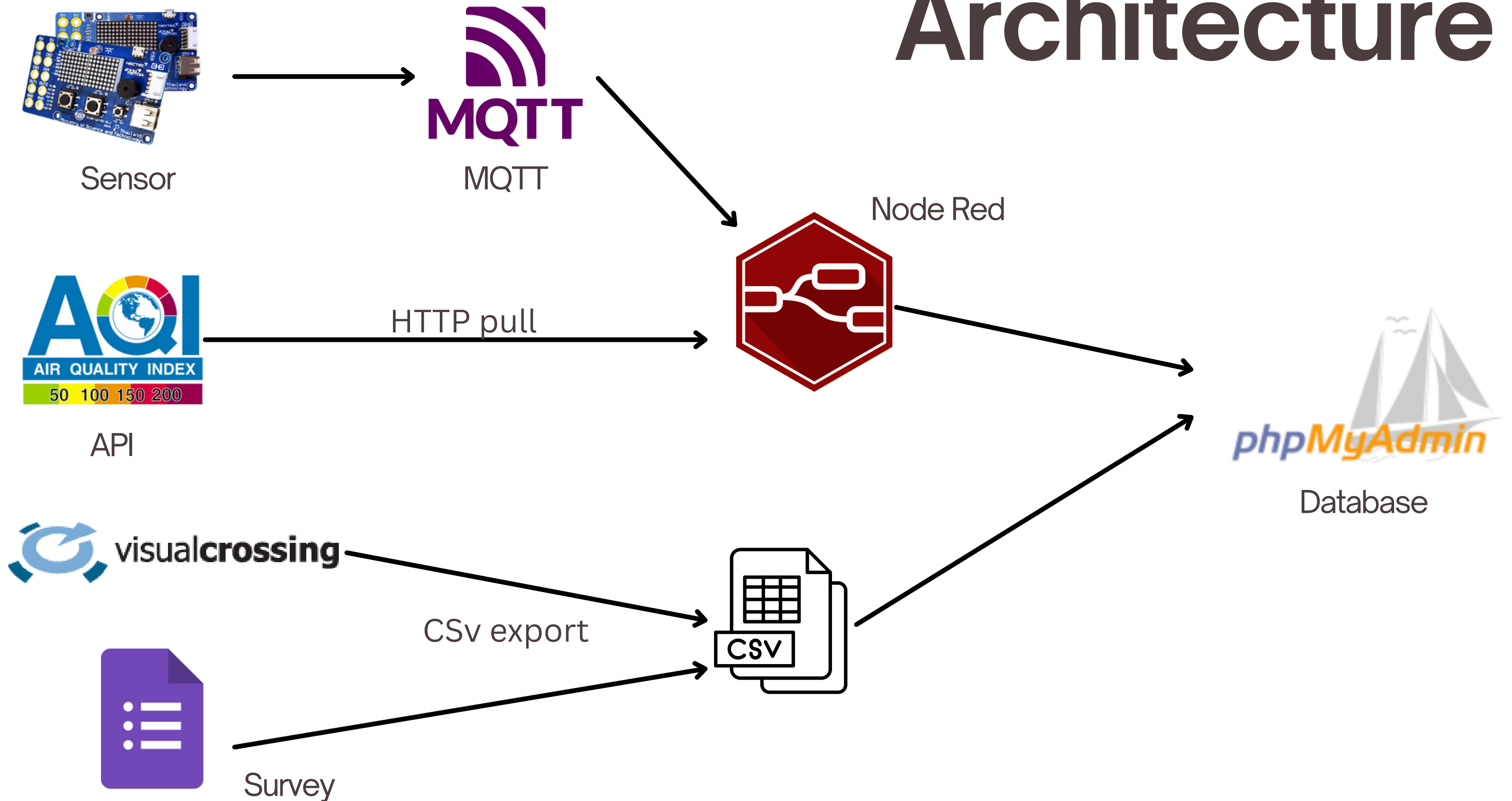
Overview

Bangkok is currently facing serious environmental challenges, including rising temperatures, air pollution (PM2.5), and unpredictable weather. Despite these conditions, people still need to travel for work and education. This raises an important question: Do environmental conditions influence people's decisions to go outside or stay indoors?

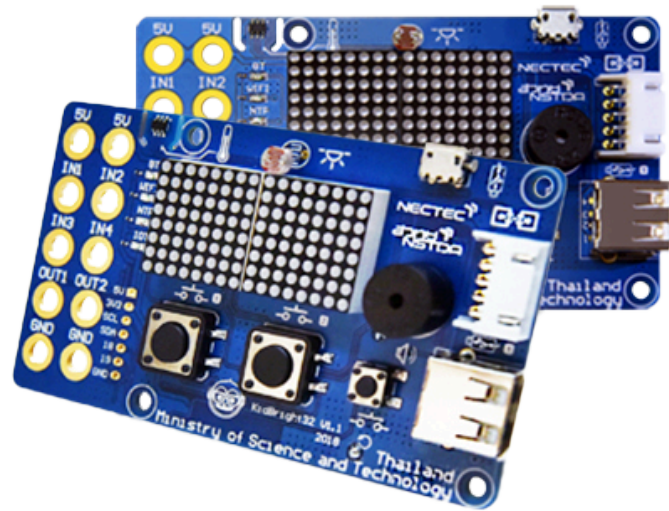
Objectives

- To evaluate the impact of weather and environmental conditions on outdoor activity decisions.
- To monitor and visualize air quality (PM2.5) and UV Index for public health safety.
- To analyze mobility patterns based on weekly travel frequency and environmental factors.

Architecture



Data Collection – Primary Source



Kidbright with sensors

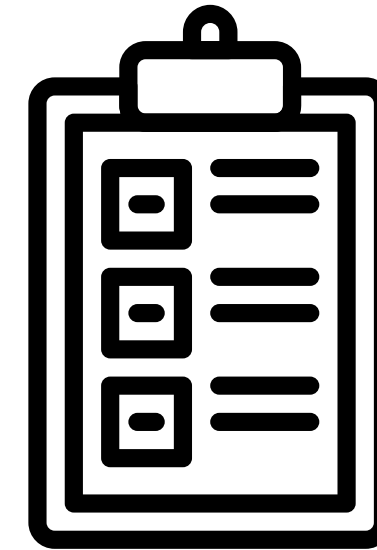
Using kidbright with these sensors

- Dust Sensor (PMS7003)
- Temperature and humidity sensor (KY-015)

MQTT

The kidbright sent data to MQTT to sent the data into database.

Data Collection – Primary Source



Survey – google form

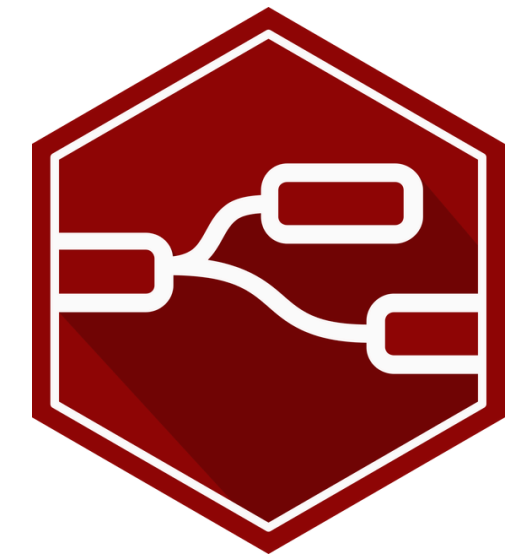
Using the google form do get the data that can be transform into sheets and cvs easily

<https://forms.gle/pDiVoAHZRTgeX4VQA>

Survey

The question is sent to friends to target their behavior of going outside.

Data Collection – Secondary Source



API

The website with data about air pollution in more than 30 districts of Bangkok, with a free API, includes crucial data like PM2.5 and air quality.

<https://aqicn.org/city/bangkok/>

API with NodeRed

Using the Node-Red to pull real time api every 2 hours, the website itself allow only 5 request per minute so every district divided into group of 5 and calling the api in different minutes to get the data smoothly

Data Collection – Secondary Source



<https://www.visualcrossing.com>

visualcrossing

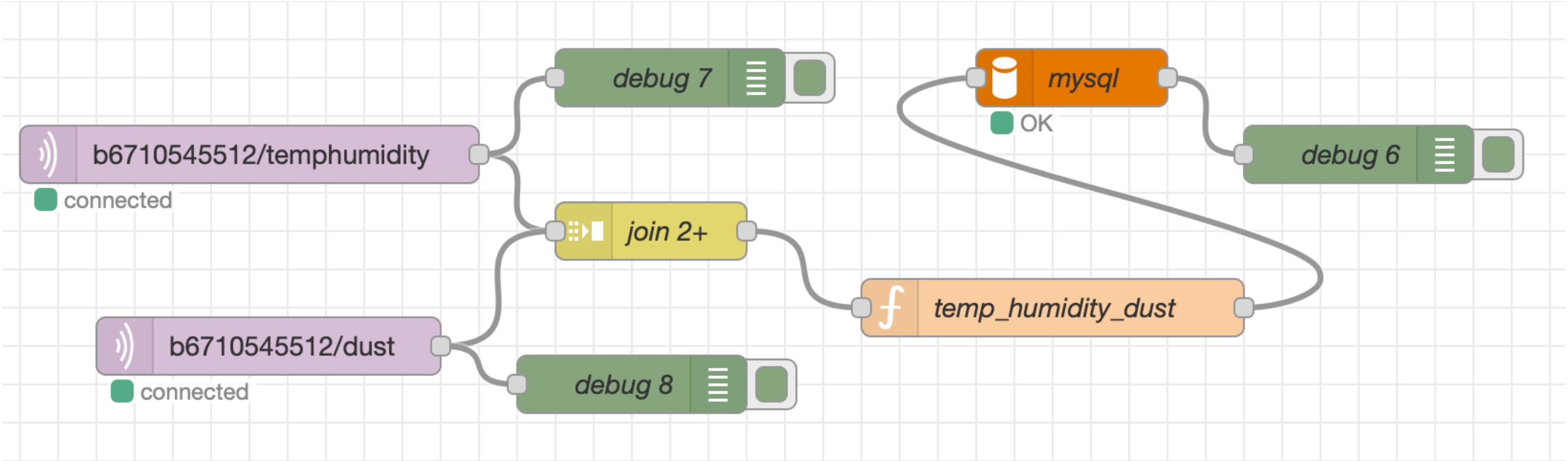
Visual Crossing provides a comprehensive weather API used by Fortune 500 companies, offering accurate historical data and real-time forecasts. It helps users convert weather data into actionable business intelligence through tools like the Query Builder and AI-assisted coding for efficient decision-making.



Data downloading with csv

The website provides the weather data of Bangkok for free. We can choose the date range we want and then get the data about weather including rain, temperature, and humidity.

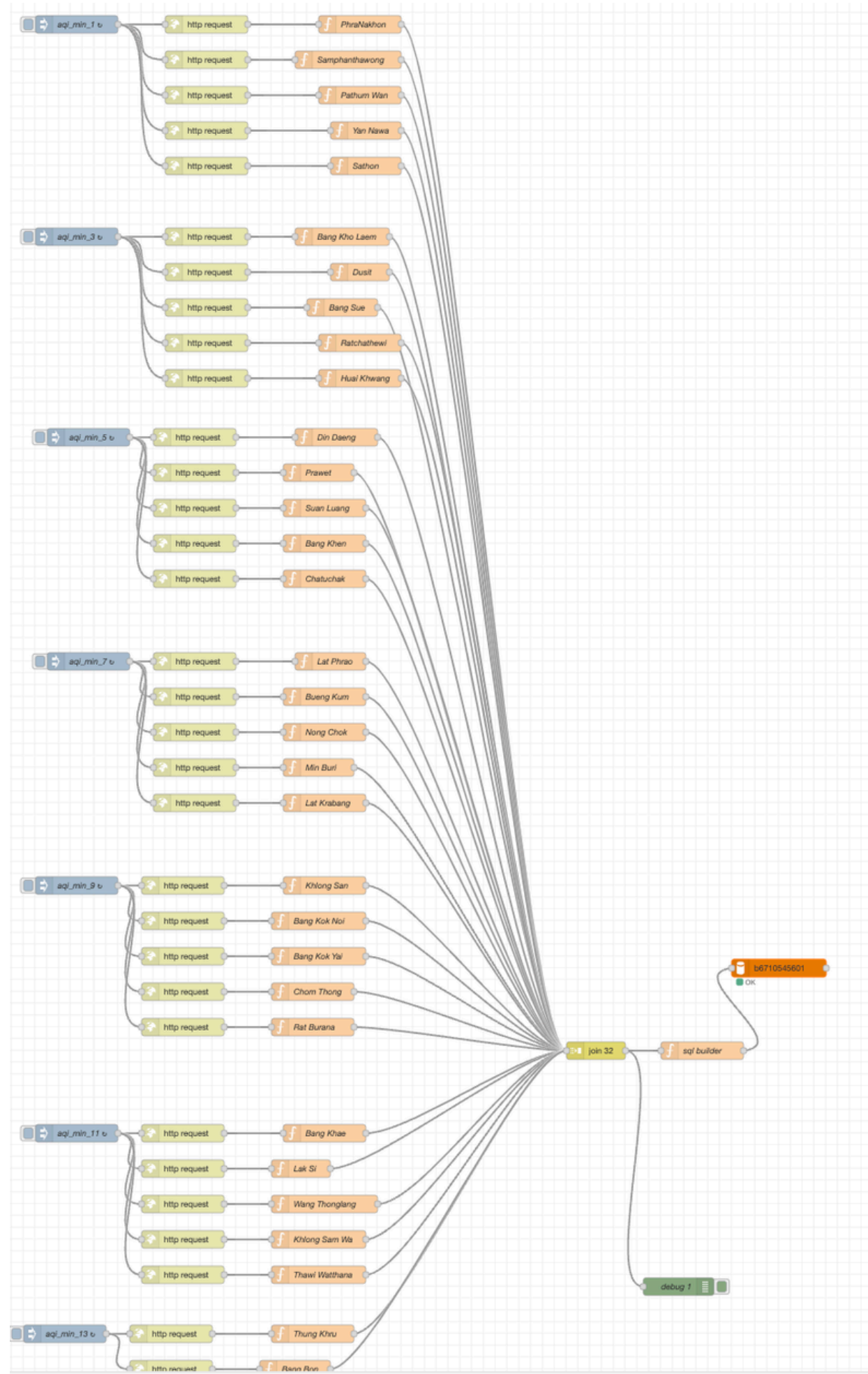
Database schema and Integration



☐	Edit	Copy	Delete	5	2026-04-21 03:18:17	25	30	51
☐	Edit	Copy	Delete	6	2026-04-21 03:18:42	16	30	51
☐	Edit	Copy	Delete	7	2026-04-21 03:19:33	16	30	55
☐	Edit	Copy	Delete	8	2026-04-21 03:19:34	34	30	55
☐	Edit	Copy	Delete	9	2026-04-21 03:29:46	20	30	61

from sensors

Database schema and Integration



Edit function node

Delete Cancel Done

Properties

Name PhraNakhon

Setup On Start On Message On Stop

```
1 let d = msg.payload.data;
2 let lat = d.city.geo[0];
3 let lon = d.city.geo[1];
4 let pm25 = d.iaqi.pm25 ? d.iaqi.pm25.v : null;
5 let aqi = d.aqi;
6
7 // Use null (the data type) instead of 'NULL' (the string)
8 if (aqi === '-' || aqi === undefined) { aqi = null; }
9 if (pm25 === undefined) { pm25 = null; }
10
11 // Output just the raw data for this district
12 msg.payload = {
13   lat: lat,
14   lon: lon,
15   district: "PhraNakhon",
16   pm25: pm25,
17   aqi: aqi
18 };
19
20 return msg;
```

Edit http request node

Delete Cancel Done

Properties

Method GET

URL https://api.waqi.info/feed/A539089/?token=a8b0C

Payload Ignore

☐ Enable secure (SSL/TLS) connection

☐ Use authentication

☐ Enable connection keep-alive

☐ Use proxy

☐ Only send non-2xx responses to Catch node

☐ Disable strict HTTP parsing

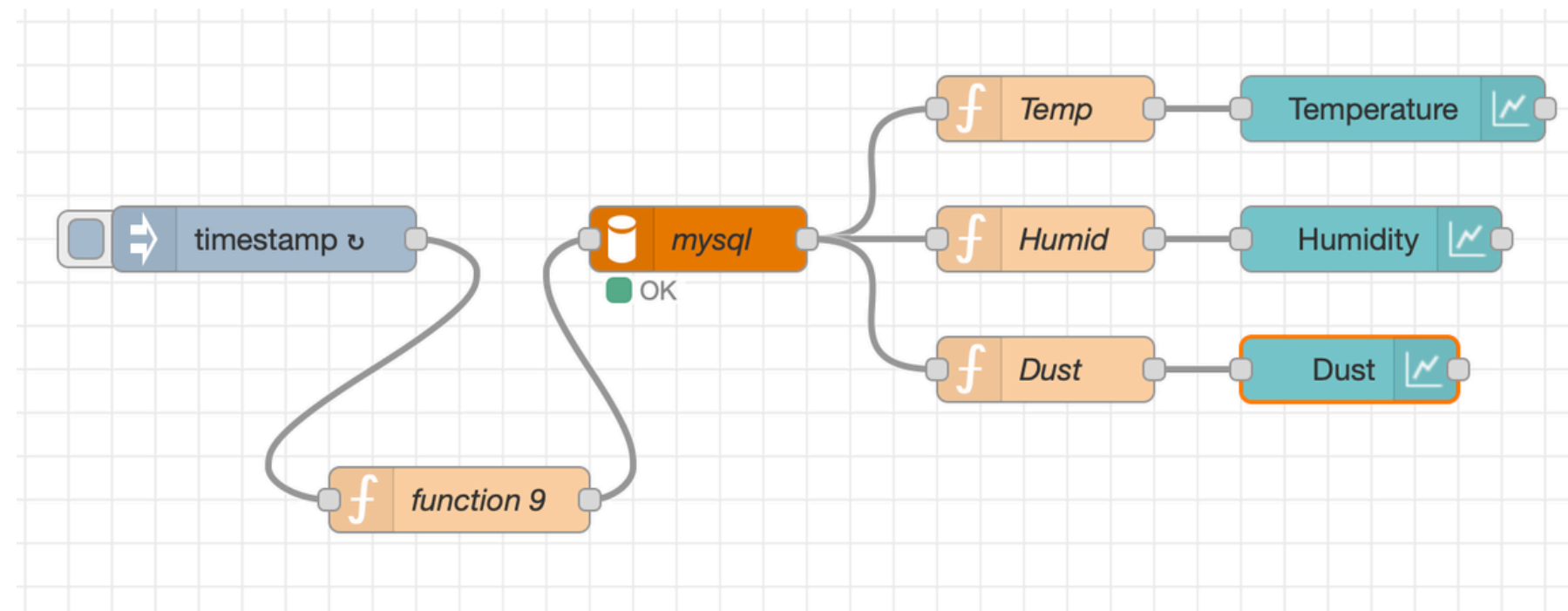
Return a parsed JSON object

Tip: If the JSON parse fails the fetched string is returned as-is.

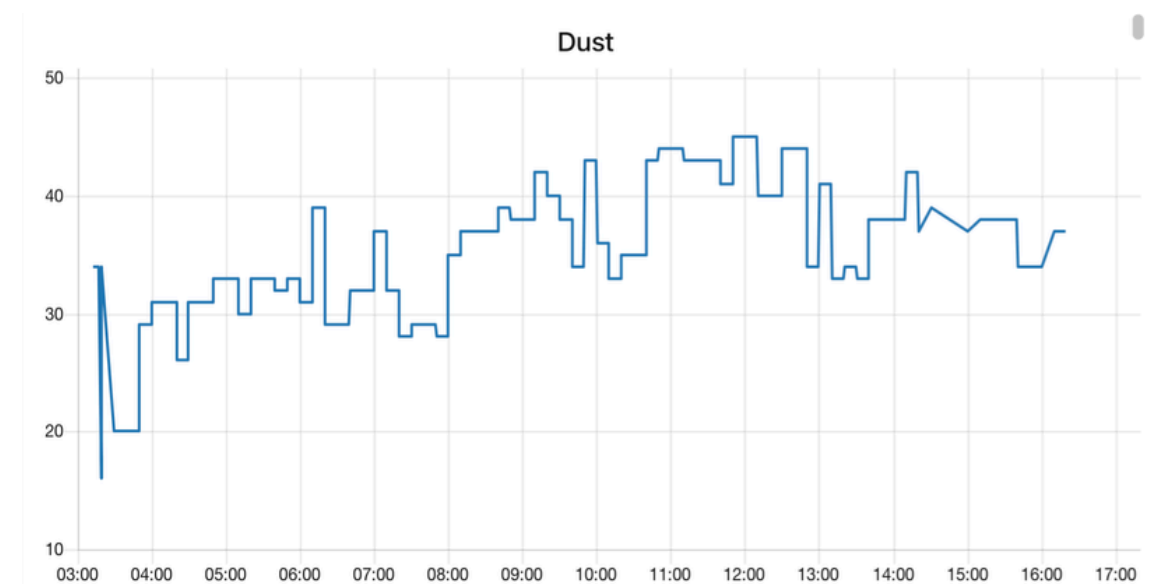
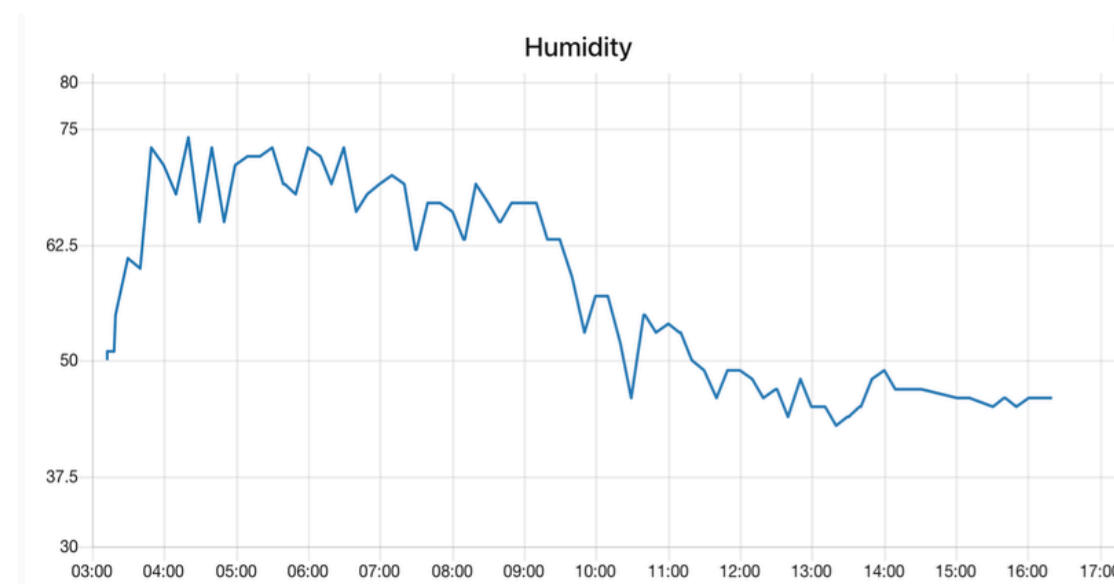
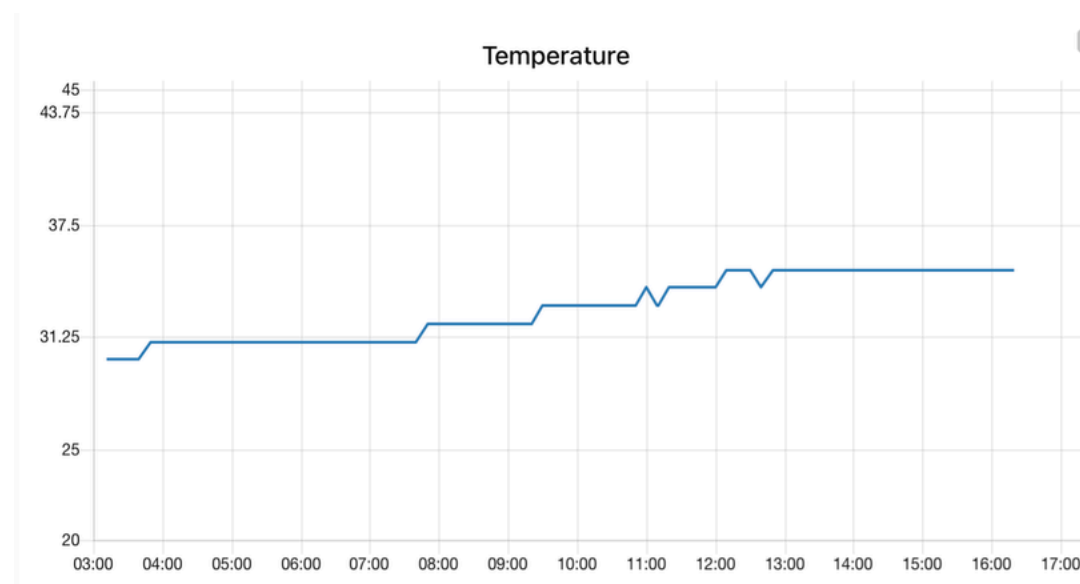
Headers

API - secondary source

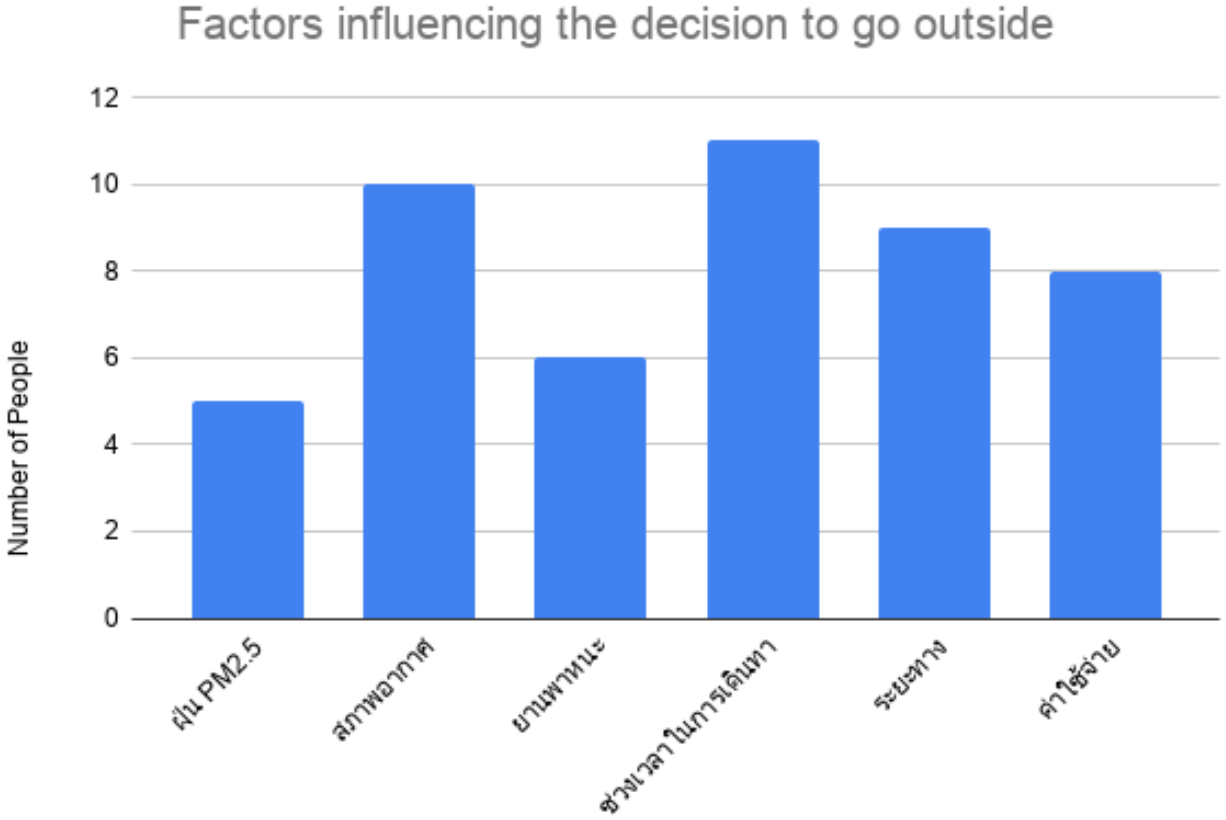
Data Visualization



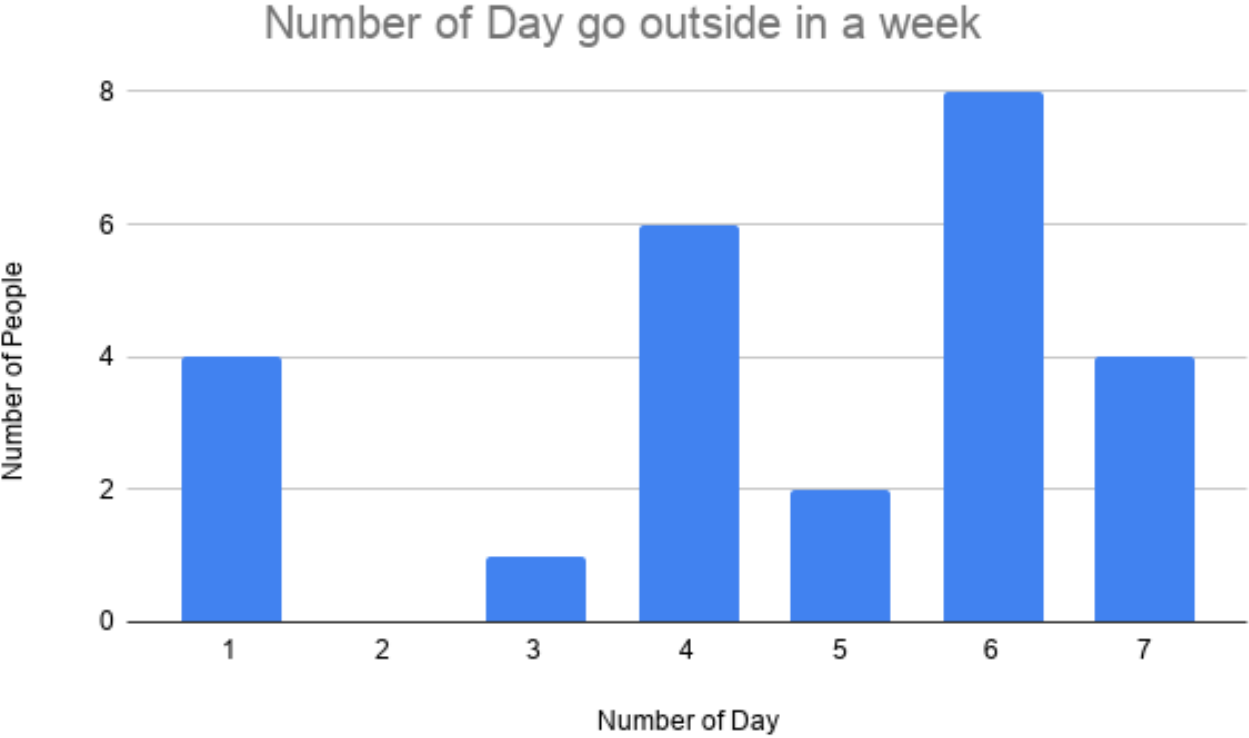
Sensors + Kidbrights



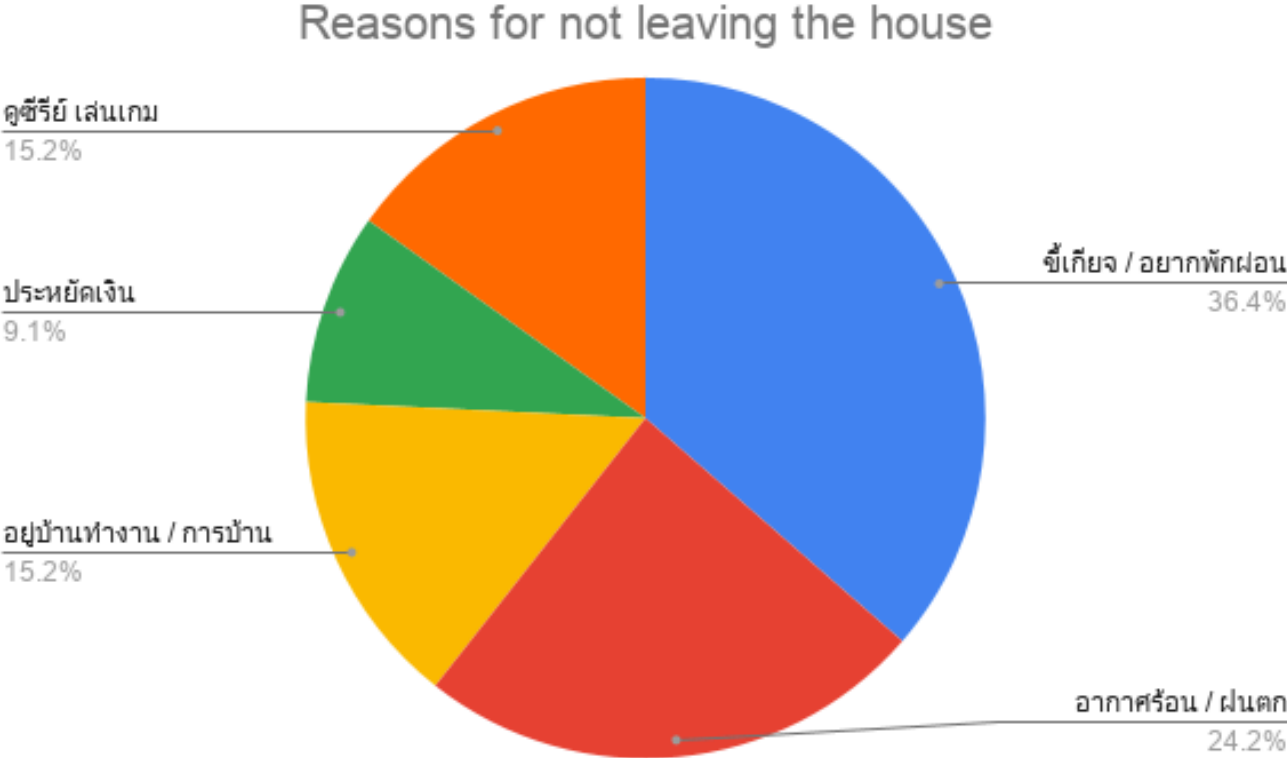
Data Visualization



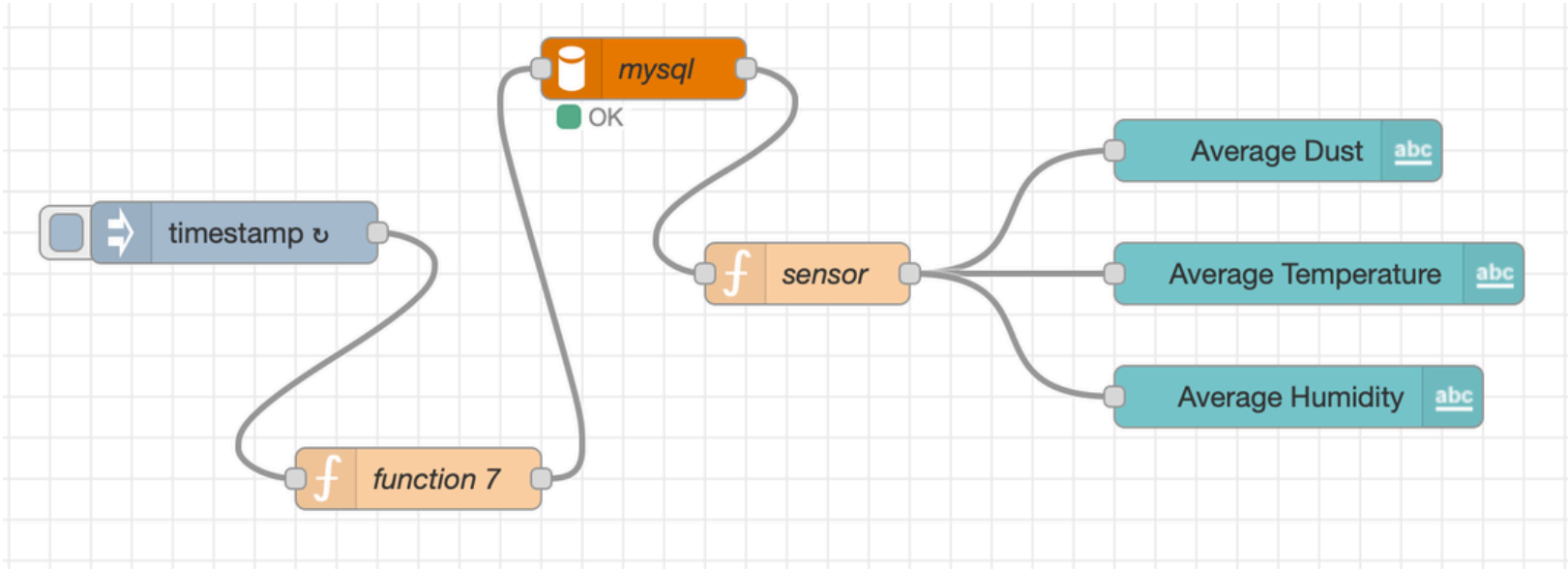
Factors influencing the decision to go outside



Surveys



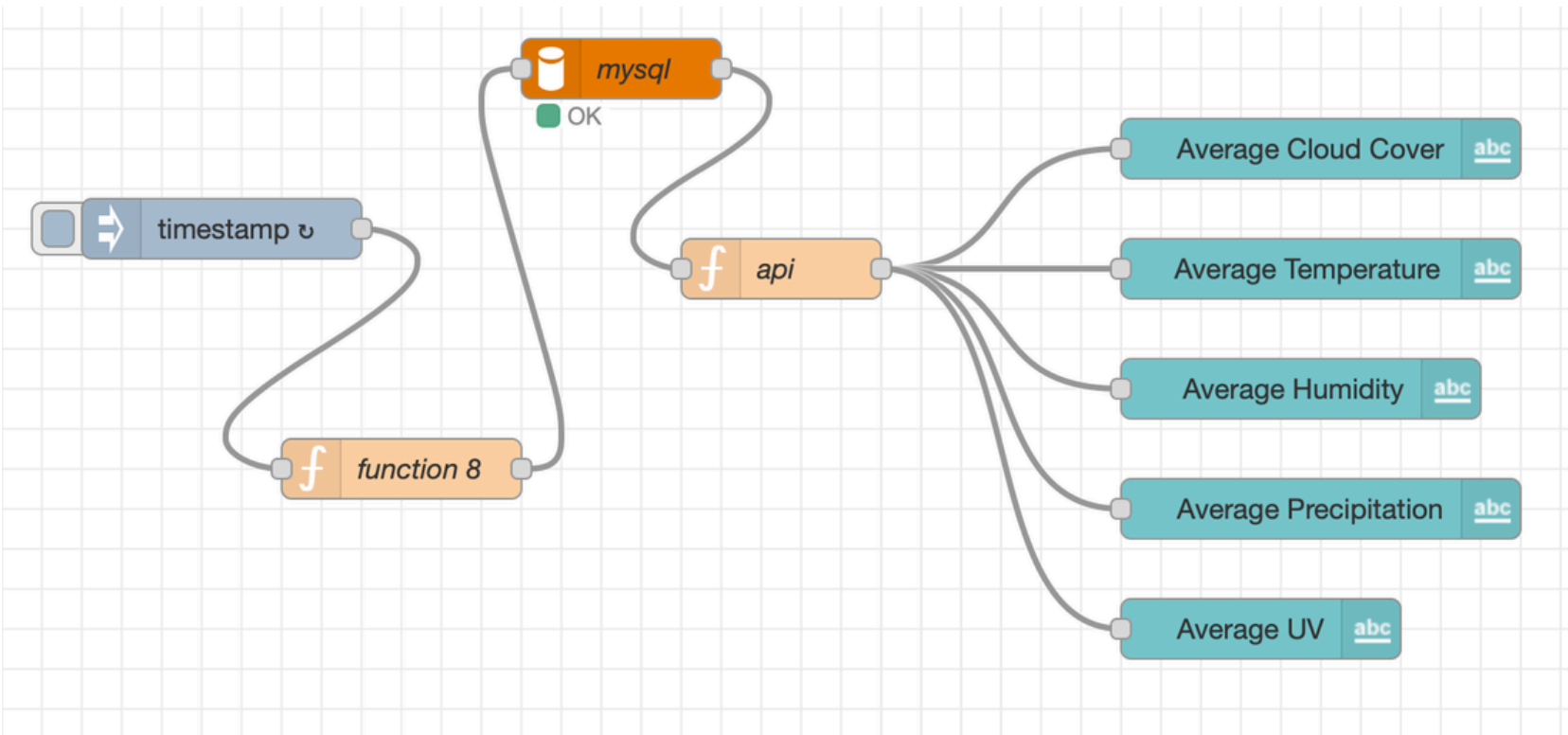
Key Features



Air Data - Sensor

Average Dust	35.0 $\mu\text{g}/\text{m}^3$
Average Temperature	32.5 $^{\circ}\text{C}$
Average Humidity	59.0 %

Sensors



BKK Weather - API

Average Cloud Cover	28.0 %
Average Temperature	32.2 $^{\circ}\text{C}$
Average Humidity	62.5 %
Average Precipitation	0.00 mm
Average UV	2.0

API



Thanks you